

Tyler Varma

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EDUCATION

The University of Georgia | Athens, GA

Aug 2026

B.S. in Computer Science | GPA: 3.73/4.0

Certificate of Applied Data Science

Relevant Coursework: Algorithms, Database Management, Data Science I & II, Cybersecurity, Computer Architecture, Data Structures, Discrete Math, Systems Programming, Software Development, Statistical Methods, Calculus I & II

TECHNICAL SKILLS/CERTIFICATIONS

Programming Languages: Python, Java, SQL, C++, C, JavaScript, R, HTML, CSS, Assembly

Frameworks & Libraries: Pandas, NumPy, Scikit-learn, PyTorch, Matplotlib, Docker, Node.js, React.js, JavaFx

Tools & Software: Git, VS Code, Jupyter Notebook, Scalation, Docker, JMP Pro, Eclipse, MS Excel, Power BI

WORK EXPERIENCE

Tutor | Tutor.com

February 2024 - Present

- Tutored over 150 students in math, statistics, and computer science through personalized one-on-one sessions.
- Introduced and explained challenging concepts providing examples for students to reference.
- Completed 100+ tutoring sessions across multiple subjects, strengthening communication and teaching skills.

Data Analysis Intern | EnDimensions

May 2019 - August 2019

- Collected, organized, and maintained competitor and partner data in Excel to support business decision-making.
- Analyzed categorical and numerical data across 10+ companies to identify trends and summarize findings clearly.

PROJECTS

Mushroom Species Recognition | Computer Vision

April 2026

- Developed and tested a computer vision pipeline in Python using CNNs and transfer learning to classify 100+ mushroom species from image data.
- Used a large multiclass dataset in Kaggle Notebook format, handling images with varying growth stages and visual conditions to support model training and evaluation.
- Contributed to the training notebook and model experimentation process, assessing classification performance with Top-1/Top-5 accuracy, precision, recall, F1-score, and confusion matrices.

Music Genre Classification | Data Science II

December 2025

- Built an end-to-end machine learning pipeline to classify Spotify tracks into 24 genres using structured audio features.
- Implemented data preprocessing steps including data cleaning, feature scaling, and hybrid class balancing to address class imbalance and missing values across genres.
- Conducted hyperparameter tuning and comparative analysis, finding that ensemble tree-based models achieved the strongest performance, with XGBoost reaching ~54% accuracy and significantly outperforming a random baseline.

Student Housing Management System | Database Management

November 2025

- Developed a full-stack student housing web application using Java, Spring Boot, Thymeleaf, MySQL, and JDBC.
- Wrote SQL queries and contributed to DDL schema design for a relational database supporting users, properties, bookings, and reviews.
- Helped implement and test core platform features including authentication, property listings, booking workflows, and review functionality.